

Last updated: August 1, 2026

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RESEARCH INTERESTS

Methods: Trustworthy AI (Explainable AI, Uncertainty Quantification), Physics-AI Hybrid Modeling, MLOps for Real-World AI Deployment.

Domains: Health Informatics & Infectious Disease Surveillance (Zoonotic & Emerging Diseases), Environmental AI (Water/Air Quality, Algal Blooms), AI for Computational Social Science & Public Policy.

EDUCATION

Korea Advanced Institute of Science and Technology (KAIST) Ph.D. in Engineering (College of Business, Graduate School of Technology and Innovation Management)	<i>Daejeon, Republic of Korea</i> Feb. 2012 – Aug. 2016
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- *Dissertation:* “Essays on big data analytics in industrial research: Green energy policy and patent litigation.”
- *Advisor:* Prof. Minki Kim

Korea Advanced Institute of Science and Technology (KAIST) Master of Engineering (College of Business, Graduate School of Technology and Innovation Management)	<i>Daejeon, Republic of Korea</i> Sep. 2009 – Aug. 2011
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Korea Advanced Institute of Science and Technology (KAIST) Bachelor of Engineering (Information and Communication Engineering; IT Business minor)	<i>Daejeon, Republic of Korea</i> Feb. 2005 – Aug. 2009
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WORK EXPERIENCE

Hankuk University of Foreign Studies <i>Associate Professor, Division of Social Science & AI</i>	<i>Seoul, Republic of Korea</i> Sep. 2025 – Present
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Tech University of Korea <i>Associate/Assistant Professor, Department of Business Administration</i>	<i>Siheung-si, Republic of Korea</i> Mar. 2017 – Aug. 2025
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- Director, SW Education Center (Sep. 2022 – Jul. 2024)
- Chief Professor, Venture Entrepreneurship Integration Major (Nov. 2019 – Sep. 2022)
- Chief Professor, Smart Factory Integration Major (Sep. 2018 – Aug. 2019)

AI Korea Inc. <i>CEO and Founder</i>	<i>Incheon, Republic of Korea</i> Dec. 2022 – Aug. 2025
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- Developed and commercialized AI prediction solutions for environmental monitoring agencies, bridging academic research and operational deployment.

Korea Environment Institute

Research Fellow, Big Data Research Team

Sejong, Republic of Korea

Jul. 2016 – Feb. 2017

PROFESSIONAL SERVICE

Korea Society for Innovation Management & Economics (KOSIME)

*Vice Chair, Special Committee on Energy Industry & Technology
Policy*

South Korea

2026 – Present

Korea Environmental Policy and Administration Society (KEPAS)

Board Member (이사)

South Korea

2026 – Present

International Telecommunications Policy Review

(정보통신정책연구, KCI)

Editorial Board Member (Management Field)

South Korea

2026 – Present

Journal of Innovation & Knowledge (SSCI, IF 14.0, JCR Top 0.5%)

Editorial Board Member

Amsterdam, Netherlands

Apr. 2025 – Present

Ministry of Environment

Committee Member, Environmental Economics Division

South Korea

Jul. 2022 – Jul. 2024

PUBLICATIONS

Journal Articles

** denotes corresponding author. Impact factors and JCR percentiles reflect the most recent JCR edition.*

- [1] **D. Lee*** (2026). “Heterogeneous transfer learning for robust infectious disease forecasting: A data-centric approach.” *Expert Systems with Applications*, 332, 133728. (SCIE, IF 9.4 / JCR Top 5.0%)
- [2] **D. Lee***, H. Jeon (2025). “Reinforced explainable AI for algal bloom forecasting under climate change: A multi-run class activation mapping (CAM) approach.” *Journal of Cleaner Production*, 529, 146805. (SCIE, IF 10.7 / JCR Top 6.7%)
- [3] M. Oh, G. Lee, **D. Lee**, J. Ahn, J. Shin (2025). “Enhancing Small-Business Ecosystems Through Local Currency Designs in South Korea: A Choice Experiment Approach.” *Systems*, 13(5), 339. (SSCI, IF 3.8 / JCR Top 6.4%)
- [4] S. Park, J. Ryu, **D. Lee**, J. Shin (2025). “How Can We Improve the Consumer Acceptance Level for Disposers Considering Regional Characteristics?” *Water*, 17(4), 493. (SCIE, IF 3.5 / JCR Top 52.0%)
- [5] **D. Lee***, B. Lee (2024). “Building Reliable AI for Quantifying Uncertainty in Particulate Matter Predictions with Deep Learning.” *Journal of Cleaner Production*, 473, 143457. (SCIE, IF 10.7 / JCR Top 6.7%)
- [6] M. Kim, **D. Lee*** (2023). “Why Uncertainty in Deep Learning for Traffic Flow Prediction Is Needed.” *Sustainability*, 15(23), 16204. (SSCI, IF 4.1 / JCR Top 34.4%)

- [7] **D. Lee***, M. Kim, B. Lee, S. Chae, S. Kwon, S. Kang (2022). “Integrated explainable deep learning prediction of harmful algal blooms.” *Technological Forecasting and Social Change*, 185, 122046. (SSCI, IF 13.5 / JCR Top 1.4%)
- [8] S. Chae, J. Shin, S. Kwon, S. Lee, S. Kang, **D. Lee*** (2021). “PM10 and PM2.5 real-time prediction models using an interpolated convolutional neural network.” *Scientific Reports*, 11(1), 11952. (SCIE, IF 4.9 / JCR Top 14.6%)
- [9] S. Chae, S. Jang, S. Lee, **D. Lee*** (2020). “Complex System Analysis of Korean Peninsula Earthquake Data.” *Scientific Reports*, 10(1), 2686. (SCIE, IF 4.9 / JCR Top 14.6%)
- [10] S. Huh, H. Lee, J. Shin, **D. Lee**, J. Jang (2018). “Inter-fuel substitution path analysis of the Korea cement industry.” *Renewable and Sustainable Energy Reviews*, 82, 4091-4099. (SCIE, IF 18.0 / JCR Top 3.1%)
- [11] J. Shin, S. Kang, **D. Lee***, B. Hong (2018). “Analysing the failure factors of eco-friendly home appliances based on a user-centered approach.” *Business Strategy and the Environment*, 27(8), 1399-1408. (SSCI, IF 13.2 / JCR Top 1.8%)
- [12] S. Chae, S. Kwon, **D. Lee*** (2018). “Predicting Infectious Disease Using Deep Learning and Big Data.” *International Journal of Environmental Research and Public Health*, 15(8), 1596. (SCIE/SSCI at time of publication; 2018 IF 2.468 / JCR Top 22.8%)
- [13] S. Lee, **D. Lee*** (2018). “Improved Prediction of Harmful Algal Blooms in Four Major South Korea’s Rivers Using Deep Learning Models.” *International Journal of Environmental Research and Public Health*, 15(7), 1322. (SCIE/SSCI at time of publication; 2018 IF 2.468 / JCR Top 22.8%)
- [14] Y. Yoon, H. Zo, M. Choi, **D. Lee**, H. Lee (2018). “Exploring the dynamic knowledge structure of studies on the Internet of things: Keyword analysis.” *ETRI Journal*, 40(6), 745-758. (SCIE, IF 2.0 / JCR Top 65.7%)
- [15] **D. Lee**, S. Kang, J. Shin (2017). “Using Deep Learning Techniques to Forecast Environmental Consumption Level.” *Sustainability*, 9(10), 1894. (SSCI, IF 4.1 / JCR Top 34.4%)
- [16] **D. Lee**, S. Kang, J. Shin (2017). “Determinants of Pro-Environmental Consumption: Multi-country Comparison Based upon Big Data Search.” *Sustainability*, 9(2), 183. (SSCI, IF 4.1 / JCR Top 34.4%)
- [17] **D. Lee**, J. Kim, J. Shin (2017). “Scale-free network analysis of big data for patent litigation cases in the United States.” *Journal of the Korean Physical Society*, 70(4), 431-435. (SCIE, IF 1.0 / JCR Top 66.5%)
- [18] **D. Lee**, M. Kim, J. Lee (2016). “Adoption of green electricity policies: Investigating the role of environmental attitudes via big data-driven search-queries.” *Energy Policy*, 90, 187-201. (SSCI, IF 9.3 / JCR Top 1.8%)
- [19] **D. Lee**, H. Lee, M. Choi (2016). “Examining the relationship between past orientation and US suicide rates: An analysis using big data-driven Google search queries.” *Journal of Medical Internet Research*, 18(2), e35. (SCIE, IF 8.2 / JCR Top 3.9%)

Book Chapters

- [1] S. Kang, **D. Lee*** (2022). “Use of artificial intelligence for predicting infectious disease.” In P. Keikhosrokiani, *Big Data Analytics for Healthcare: Datasets, Techniques, Life Cycles, Management, and Applications*, Elsevier. ISBN: 9780323985161 (SCOPUS).

Research Reports

- [1] **D. Lee**, J. Shin, M. Kim (2025). “Predicting the Spread of Avian Influenza in Wild Birds (II).” Research Report, NIWDCP.
- [2] **D. Lee**, J. Shin, M. Kim (2024). “Predicting the Spread of Avian Influenza in Wild Birds (I).” Research Report, NIWDCP.
- [3] S. Kang, **D. Lee**, G. Ko, D. Jin, H. Hong, D. Kim, S. Kang (2023). “Big Data Analysis: Application to Environmental Research and Service VII.” Policy Research Report, Korea Environment Institute.
- [4] S. Kang, **D. Lee**, J. Pyo, G. Ko, Y. Cho, G. Han, D. Jin, S. Kang (2022). “Big Data Analysis: Application to Environmental Research and Service VI.” Policy Research Report, pp. 1-108, Korea Environment Institute.
- [5] H. Hong, **D. Lee**, G. Ko, D. Jin, S. Kang, S. Kang, D. Kim (2021). “Big Data Analysis: Application to Environmental Research and Service V.” Policy Research Report, pp. 1-195, Korea Environment Institute.
- [6] S. Kang, **D. Lee**, G. Ko, D. Jin, H. Hong, G. Han, S. Kang, D. Kim (2019). “Big Data Analysis: Application to Environmental Research and Service III.” Policy Research Report, pp. 1-288, Korea Environment Institute.
- [7] S. Kang, **D. Lee**, K. Jang, D. Jin, H. Hong, G. Han, J. Kim, S. Kang, D. Kim (2018). “Big Data Analysis: Application to Environmental Research and Service II.” Policy Research Report, pp. 1-271, Korea Environment Institute.
- [8] S. Kang, **D. Lee**, K. Jang, D. Jin, H. Hong, G. Han, J. Kim, S. Kang, D. Kim, E. Jung (2017). “Big Data Analysis: Application to Environmental Research and Service.” Policy Research Report, pp. 1-465, Korea Environment Institute.
- [9] D. Kim, S. Song, S. Park, J. Kim, **D. Lee**, S. Jo (2017). “A Study on Expansion of Pro-Environmental Consumption Using Social Economy.” Policy Research Report, pp. 1-246, Korea Environment Institute.
- [10] S. Kwak, J. Shin, **D. Lee**, S. Kang (2016). “A Study on Promoting Environmental Consumption by Developing the Environmental Attitude-Behavior Model.” Policy Research Report, pp. 1-226, Korea Environment Institute.

Conferences (Domestic)

- [1] “On-Device Agentic AI Framework for Predicting Spatial Jumps of Emerging Infectious Diseases: A Lumpy Skin Disease (LSD) Case Study” (On-Device Agentic AI 프레임워크를 활용한 신종 감염병의 공간적 점프 예측: 럽피스킨병 (LSD) 사례 연구). *Korean Society of Zoonoses* (인수공통감염병학회), Oct. 2025.
- [2] “Explainable AI-Based Avian Influenza Outbreak Prediction Considering Uncertainty” (불확실성을 고려한 설명 가능한 인공지능 기반 조류 인플루엔자 발생 예측 연구). *Joint Conference on Economics*, Feb. 2025.
- [3] “Analysis of Industrial Workforce Market Trends and Demand Using Big Data and Large Language AI Models” (채용 빅데이터와 대형 언어 인공지능 모델을 활용한 산업 인력 시장 동향 및 수요 분석). *Summer Conference of the Korea Technology Innovation Society*, Aug. 2024.

- [4] “Spatiotemporal Modeling and Factor Analysis for Predicting Avian Influenza Spread” (조류 인플루엔자 확산 예측을 위한 시공간 모델링 및 영향 요인 분석). *Annual Conference of the Korea Environmental Economics Association*, Aug. 2024.
- [5] “Automation of Algal Bloom Prediction Through a Deep Learning-Based MLOps Pipeline: AI Model Serving for Policy Utilization” (딥러닝 기반 MLOps 파이프라인을 통한 녹조 예측 자동화: 정책적 활용을 위한 인공지능 모델 서빙 접근). *Summer Conference of the Korea Technology Innovation Society*, Jul. 2024.
- [6] “(Session for Emerging Researchers) Development and Application of AI for Environmental and Infectious Disease Predictions” ((신진연구자 세션) 환경 및 감염병 예측을 위한 인공지능의 개발과 실제). *Korea Software Congress*, Dec. 2023.
- [7] “Explainable Deep Learning Integration for Algal Bloom Prediction and MLOps Implementation” (설명 가능한 녹조 통합 딥러닝 예측과 MLOps). *Spring Conference of the Korean Society of Analytical Science*, May 2023.
- [8] “Fine Dust Prediction Using Convolutional Neural Network Deep Learning” (컨벌루션 신경망 딥러닝을 활용한 미세먼지 예측). *Summer Conference of the Korea Environmental Economics Association*, Aug. 2018.
- [9] “Stock Price Prediction and Profitability Evaluation Using Deep Learning Analysis of Financial Big Data” (재무 빅데이터의 딥러닝 분석을 통한 주가예측 및 수익률 평가). *Summer Conference of the Korea Technology Innovation Society*, Jul. 2018.
- [10] “Risk Prediction for Fine Dust Using Deep Learning Technology” (딥러닝 기술을 활용한 미세먼지 리스크 예측). *Summer Conference of the Korea Technology Innovation Society*, Jun. 2017.
- [11] “Complex System Analysis of Earthquake Big Data: Focusing on Regional Concentration Phenomena” (지진 빅데이터를 활용한 복잡계 현상 분석: 지진의 지역 집중화 현상을 중심으로). *Summer Conference of the Korea Regional Information Society*, Jun. 2017.
- [12] “Application of Deep Learning Technology in Environmental Research: Predicting Eco-Friendly Consumption Indices” (딥러닝 기술의 환경 연구에의 적용: 친환경 소비 지표 예측을 중심으로). *Summer Conference of the Korea Technology Innovation Society*, Jun. 2016.
- [13] “A Farewell to Arms? Patent War in the Smartphone Industry.” *Joint Economics Conference*, Feb. 2015.

PATENTS

- [1] “**Method for Predicting Air Quality Using Artificial Intelligence.**” Patent Application No. 10-2022-0015727, Feb. 2022. (Filed)
- [2] “**Improved Device and Method for Predicting Fine Dust Concentration.**” Patent No. 10-2022-0082928, Jul. 2022. (Registered)
- [3] “**Method and Device for Predicting Infectious Diseases Using Artificial Intelligence.**” Patent Application No. 10-2022-0132078, Oct. 2022. (Filed)
- [4] “**Method and Device for Predicting Algal Bloom Occurrences Using an Artificial Intelligence Model.**” Patent Application No. 10-2022-0172370, Dec. 2022. (Filed)
- [5] “**Method and Device for Modeling Generation for Time Series Prediction.**” Patent Application No. 10-2023-0010001, Jan. 2023. (Filed)

PROJECTS

Current Projects

- [1] **Institute for Information & Communications Technology Planning & Evaluation (IITP)** *Apr. 2026 – Dec. 2033*
Principal Investigator, “Development of an Ultra-Fast, High-Reliability Physics-AI Hybrid Disaster Response Technology Based on MambaFlow Matching” (Digital Columbus Project)
- [2] **National Research Foundation of Korea (NRF)** *Mar. 2026 – Feb. 2029*
Principal Investigator, “Development of Physics-AI Integrated Dynamic Graph Meta-Learning Based Algal Bloom Prediction Technology for Overcoming Data Scarcity” (Outstanding Young Scientist Grant)

Past Projects

- [3] **National Wildlife Disease Control and Prevention Agency** *Aug. 2025 – May 2026*
Co-Investigator, “Wild Bird Avian Influenza Spread Prediction Research(III)”
- [4] **National Wildlife Disease Control and Prevention Agency** *Jul. 2024 – May 2025*
Principal Investigator, “Wild Bird Avian Influenza Spread Prediction Research(II)”
- [5] **National Research Foundation of Korea** *Mar. 2020 – Feb. 2025*
Principal Investigator, “Simulation of Infectious Disease Prediction Using Transfer Learning Deep Learning”
- [6] **Tech University of Korea** *Oct. 2023 – Oct. 2024*
Principal Investigator, “Development of Advanced Algal Bloom Prediction Model Using Explainable AI” (1st Place, Research Competition)
- [7] **Office of Strategic R&D Planning** *Jul. 2024 – Sep. 2024*
Principal Investigator, “Utilizing Big Data and Large Language Models for Analyzing Trends and Demand in the Industrial Workforce Market”
- [8] **National Wildlife Disease Control and Prevention Agency** *Jul. 2023 – May 2024*
Principal Investigator, “Wild Bird Avian Influenza Spread Prediction Research”
- [9] **National Research Foundation of Korea** *Nov. 2023 – Mar. 2024*
Co-Investigator, “Development of Performance Indicators and Management System for LAMP Project”

- [10] **Kookmin University** *Oct. 2023 – Dec. 2023*
Principal Investigator, “Development of ML-Based Performance Prediction Model for Seawater Desalination Plants”
- [11] **Kookmin University** *Sep. 2023 – Nov. 2023*
Principal Investigator, “Development of ML-Based Prediction Model for Water Quality and Energy Consumption in Sewage Treatment Plants”
- [12] **Korea Test Laboratory** *Jun. 2023 – Oct. 2023*
Principal Investigator, “Test Data Analysis for Remaining Life Measurement of Electric Vehicle Lithium-Ion Batteries”
- [13] **Korea Environment Institute** *Mar. 2023 – Oct. 2023*
Principal Investigator, “Construction of Automated Data Pipeline for Algal Bloom Prediction Using Deep Learning for MLOps”
- [14] **National Wildlife Disease Control and Prevention Agency** *Aug. 2022 – Nov. 2022*
Principal Investigator, “Establishment of a Rapid Response System for Avian Influenza”
- [15] **Korea Test Laboratory** *Jun. 2022 – Oct. 2022*
Principal Investigator, “Research on Application of Functional Safety Standards in HVAC Using AI”
- [16] **Korea Environment Institute** *Apr. 2022 – Oct. 2022*
Principal Investigator, “Development of a New Fine Dust Prediction Module Based on Explainable AI”
- [17] **Exabyte Inc.** *Feb. 2022 – Jul. 2022*
Principal Investigator, “Development of Object Detection Model Using Deep Learning”
- [18] **Korea Environment Institute** *Mar. 2021 – Nov. 2021*
Principal Investigator, “Deep Learning-Based Prediction of Fine Dust Concentration and Uncertainty”
- [19] **Korea Design Promotion Institute** *Jun. 2020 – Dec. 2020*
Co-Investigator, “Support for Multidisciplinary Education Courses Led by Design”
- [20] **National Research Foundation of Korea** *Mar. 2017 – Feb. 2020*
Principal Investigator, “Application of Deep Learning Technology in Environmental and Industrial Research”
- [21] **Korea Environment Institute** *Feb. 2019 – Nov. 2019*
Principal Investigator, “Deep Learning-Based Integrated Prediction of Environmental Pollution”

- [22] **Jeongseok Logistics Academic Foundation** Jan. 2019 – Dec. 2019
Principal Investigator, “Deep Learning-Based Traffic Prediction for Logistics Optimization”
- [23] **Tech University of Korea** Jul. 2018 – Nov. 2018
Principal Investigator, “Quantitative Analysis of Factors Influencing Dropout Rates Based on Big Data”
- [24] **National Research Foundation of Korea** Mar. 2018 – Feb. 2019
Principal Investigator, “Application of Deep Learning Technologies — Initial Innovation Laboratory”
- [25] **Korea Environment Institute** Mar. 2018 – Nov. 2018
Principal Investigator, “Fine Dust Prediction Using Convolutional Neural Networks”
- [26] **Korea Industrial Technology Promotion Agency** Mar. 2018 – Feb. 2022
Co-Investigator, “Smart Factory Operation Design and Professional Personnel Training Project”
- [27] **National Research Foundation of Korea** Mar. 2018 – Aug. 2019
Co-Investigator, “LINC+ Leading University for Innovation in the 4th Industrial Revolution”
- [28] **Korea Environment Institute** Jul. 2017
Principal Investigator, “Natural Language Analysis and Advisory of Water Source Related Complaint Data”
- [29] **National Research Foundation of Korea** Jul. 2017 – Jun. 2019
Principal Investigator, “Predicting Epidemic Spread Using Deep Learning Analysis of Query Big Data”
- [30] **Korea Environment Institute** Apr. 2017 – Sep. 2017
Principal Investigator, “Analysis of Eco-Friendly Consumption Using Social Networking Services (SNS)”
- [31] **Tech University of Korea** Apr. 2017 – Mar. 2019
Principal Investigator, “Application of Deep Learning Technology in Social Science Research”
- [32] **Korea Environment Institute** Mar. 2017 – Nov. 2017
Principal Investigator, “Environmental Risk Prediction Using Deep Learning”

TEACHING EXPERIENCE

GIE = Global Instructor Evaluation (course evaluation score).

Hankuk University of Foreign Studies
Associate Professor

Seoul, Republic of Korea
Sep. 2025 – Present

- **Fundamentals of Social Data Programming** (M04103201). Fundamental concepts of programming using Python: data types, operators, conditionals, loops, functions, classes, modules, packages, exception handling. Fall 2025: M04103201 (GIE: 4.8/5, Univ. Avg: 4.61).
- **Industrial Data Visualization** (M04111101). Data preprocessing with Python (Numpy, Pandas); visualization with Matplotlib, Streamlit, Github for interactive dashboard. Fall 2025: M04111101 (GIE: 4.7/5, Univ. Avg: 4.61).
- **Understanding Technology Development Research** (M04112101). Technology management and high-tech industry; technology development methodologies (consumer preference, market analysis); commercialization of new technologies. Fall 2025: M04112101 (GIE: 4.23/5, Univ. Avg: 4.61).
- **Object-Oriented Programming** (M04107101). Principles and practical applications of Object-Oriented Programming (OOP) through Python; object-oriented design skills for social science data modeling. Spring 2026.
- **Deep Learning and Applications** (M04114101). Theory and practice of artificial neural networks from individual neurons to modern deep learning architectures (CNN, LSTM, etc.); implementing core mechanisms from scratch with NumPy, then building practical models with PyTorch for data analysis. Spring 2026.

Tech University of Korea
Associate/Assistant Professor

Siheung-si, Republic of Korea
Mar. 2017 – Aug. 2025

11 distinct courses across undergraduate and graduate levels (60+ sections). Consistently above department average. Representative courses:

- **Big Data Analysis** (R programming, statistics, visualization). 14 sections over 7 years. Best GIE: 4.77/5.
- **Fundamentals of Programming** (Java). 8 years. Peak GIE: 4.99/5 (Spring 2023).
- **Programming and Data Analysis** (Python, OOP, data preprocessing). 6 semesters.
- **Advanced IT Service Techniques** (Deep learning/ML with Python). 5 semesters. GIE: 4.73/5.
- **Graduate: Information Strategy and Management Innovation** (MIS, big data, AI). 2 semesters.
- **Graduate: New Technology Management** (Technology management theories). 1 semester.
- **Management Capstone Design I/II/Planning** (Undergraduate research). 6 years.

KOCW (Korea OpenCourseWare)
Official Public Online Course

South Korea, kocw.net

- **Python Programming**. Fundamental programming with Python (conditionals, loops). Online since Spring 2020. 26 lectures. GIE: 4.4/5, Views: 41,000. [Link](#).
- **Python Programming 2**. Object-oriented Python, Numpy, Pandas. Online since Fall 2020. 26 lectures. Views: 19,000. [Link](#).
- **Big Data Analysis**. R programming and data analysis for big data. Online since Fall 2019. 26 lectures. GIE: 4.5/5, Views: 28,000. [Link](#).
- **Fundamentals of Programming**. JAVA, core programming concepts. Online since Spring 2020. 26 lectures. GIE: 5.0/5, Views: 11,000. [Link](#).

K-DIGITAL Training Online Course
Instructor

Siheung-si, Republic of Korea

- **Deep Learning**. Principles of deep learning; implementing models with Numpy and Python (without Tensorflow/PyTorch). Jun. 2021 – Aug. 2023 (Cohorts 1–6). 20 lectures.

HONORS & AWARDS

- **President's Award of the Korea Environment Institute** (한국환경연구원 장상) (2024)
Environmental Data Analysis Contest: Development of Multimodal LLM Interactive AI
- **Outstanding Researcher Award (Excellence in Academic Paper Category)** (2019)
With the President's Commendation at Tech University of Korea
- **Presidential Award for the Korea Young Talent Award** (대한민국 인재상 대통령상) (2008)
Presented by the President of South Korea

TECHNICAL COMPETENCIES

- **Programming:** Python, R, Java, SQL
- **AI/ML Frameworks:** PyTorch, TensorFlow, scikit-learn, NumPy, Pandas
- **Statistical Tools:** SPSS, Stata
- **MLOps & Infrastructure:** Docker, Streamlit, GitHub Actions, REST API deployment
- **Visualization:** Matplotlib, Streamlit dashboards, interactive web applications